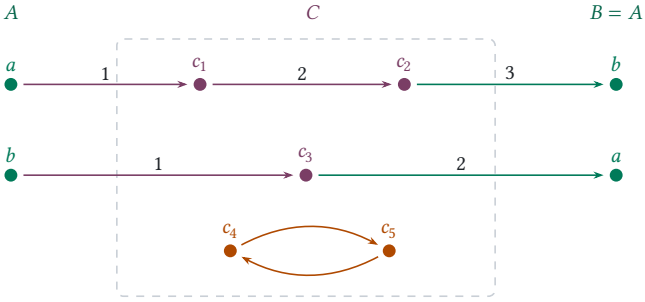
(a) One application of f

$$A \sqcup C \xrightarrow{f} B \sqcup C$$

$$\begin{aligned} a &\mapsto c_1 \\ b &\mapsto c_3 \\ c_1 &\mapsto c_2 \\ c_2 &\mapsto b \\ c_3 &\mapsto a \\ c_4 &\mapsto c_5 \\ c_5 &\mapsto c_4 \end{aligned}$$

(b) Feed only C back



$$\underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}}_{x_0} + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{x_1} + \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{x_2} = \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\mathrm{Tr}^C f}$$